

TEAMING AGREEMENT

This Teaming Agreement ("Agreement") is made as of the last signature date below ("the Effective Date") by and between **Wipro Limited**, a public limited liability company having its registered office at **Doddakannelli, Sarjapur Road, Bangalore 560035, India** ("WIPRO") and **nCircle Tech Pvt. Ltd**, registered office at **Unit-2, Office no. 3, 4th Floor, IT -7, Qubix (SEZ), Blue Ridge, Hinjewadi, Phase-1, Pune-411057** a company incorporated under the laws of India ("Company"). Each of WIPRO and Company may be referred to as a "Party" or collectively as "Parties".

WHEREAS

- A. WIPRO is engaged in the information technology and information technology enabled services, and the Company is involved in offering AI-driven 3D visualization and CAD-BIM solutions.
- B. The Parties desire to combine their complementary skills and expertise in the preparation of proposals ("Proposals" or "Proposal") to be submitted to identified third party customers ("Customers") in response to Customers' Request for Proposal for a project or other similar joint bids for work submitted to Customers ("Work").
- C. The respective Proposal will designate WIPRO as the prime contractor ("Prime Contractor") and the other Party as the subcontractor ("Subcontractor") for the products and services to be provided.
- D. If a Proposal is accepted, the Prime Contractor will enter into a Prime Contract. Pursuant to the Prime Contract, the Prime Contractor and the Subcontractor shall enter into a statement of work governed by this Agreement in relation to the Work under the accepted Proposal ("**Sub Contract**" / "**Statement of Work**" / "**SOW**" in **Appendix - 1**).

NOW IT IS HEREBY AGREED AS FOLLOWS:

1 DEFINITIONS

- 1.1. "**Work Description**" shall mean and refer to the form of work description set out as an exhibit to this agreement that will carry the details of the work that the Parties shall undertake pursuant to this Agreement in relation to each Proposal.
- 1.2. "**Prime Contract**" means the agreement which the Prime Contractor shall enter into with the Customer for the provision of services if the Proposal is accepted by the Customer.
- 1.3. "**Proposal**" shall have the meaning assigned to the term under Recital B hereinabove.
- 1.4. "**Request**" means the request for proposal or invitation to bid issued by Customer with respect to a Project.
- 1.5. "**Subcontract**" shall have the meaning assigned to the term under Recital B hereinabove.
- 1.6. "**Work**" shall have the meaning assigned to the term under Recital B hereinabove.

2 SCOPE OF AGREEMENT

- 2.1. The Parties agree and undertake to cooperate in the preparation of Proposals to be submitted by the Prime Contractor.
- 2.2. For every Proposal to be submitted, Parties will formulate, agree and sign a Work Description. Each Party agrees to undertake the respective activities set forth in this Agreement and in the Work Description.

3 OBLIGATIONS OF THE PARTIES

3.1. General.

- 3.1.1 The role and contribution of each Party in relation to the Work will be described in general terms in the Work Description.
- 3.1.2 Each Party shall appoint a representative to supervise and co-ordinate its performance of its obligations under this Agreement and under each Proposal that is accepted and Prime Contract executed pursuant to the Proposal. The representative shall provide professional and prompt liaison with the other Party and have the necessary expertise and authority to make a commitment on behalf of the appointing Party.
- 3.1.3 All contact and communication with Customer pertaining to the Request, Proposal and Prime Contract shall be coordinated through Prime Contractor. The Subcontractor agrees to promptly notify the Prime Contractor if it is directly contacted by Customer concerning the Request, Proposal, Prime Contract or any related matter.
- 3.1.4 Unless otherwise agreed by the Parties in writing, each Party shall bear its own costs and expenditures incurred in connection with the preparation, submission and negotiation of the Proposal. Neither Party shall be liable for the costs incurred or other obligations undertaken by the other Party in connection with the Proposal or any such negotiation.
- 3.1.5 Subcontractor and its representatives shall make themselves available for consultation with Prime Contractor during any negotiations with Customer. The subcontractor shall, upon Prime Contractor's request, attend any negotiations or discussions between Prime Contractor and Customer which pertain to the Proposal or Prime Contract. However, WIPRO shall not be obligated to involve the Subcontractor in such discussions or negotiations.

3.2. Preparation and Submission of the Proposal.

- 3.2.1 Prime Contractor will furnish Subcontractor with the relevant portion of the Request, including any terms and conditions required by Customer to be incorporated into the Prime Contract. Upon receipt of the Request, Prime Contractor and Subcontractor shall agree on the format of Subcontractor's inputs and the time frame for its submission to Prime Contractor.
- 3.2.2 Prime Contractor shall be responsible for the preparation, content, evaluation and submission to Customer of the Proposal and Prime Contract. During the preparation of the Proposal, Subcontractor will be provided with an opportunity to review the areas of the Proposal relating to the portions of the Work to be supplied by Subcontractor. The Subcontractor acknowledges that PRIME CONTRACTOR retains ultimate control over the form and content of the Proposal.

- 3.2.3 Each Party shall draw up at its own cost a proposal with respect to its responsibilities detailed in the applicable Work Description and as required by the Request. Each Party shall be solely responsible for its portion of the Proposal, including the accuracy and adequacy of designs, interface and technical data, appropriate performance parameters, support and all other matters proposed by it and for the completion and delivery times for the Work proposed by it in its portion of the Proposal. In addition, Subcontractor will provide PRIME CONTRACTOR with any exceptions, in good faith, it may have to the terms and conditions required by the Request to be incorporated into the Subcontract and Prime Contract. During preparation of their respective proposals, the Parties shall meet regularly to review and discuss its progress and resolve any issues.
- 3.2.4 In the event that the Request calls for the Work of Subcontractor and PRIME CONTRACTOR to be compatible, the Parties shall consult with one another and ensure that their respective proposals meet such requirements.
- 3.2.5 Subcontractor shall submit to PRIME CONTRACTOR its proposal, including prices, at the agreed time and in the agreed format, in the form of an irrevocable offer valid until the later of: (i) last date for acceptance of the Proposal by the Customer, or any extension thereto agreed by PRIME CONTRACTOR and Customer; and (ii) 180 days from the date of submission of the proposal to PRIME CONTRACTOR.
- 3.2.6 Subcontractor shall identify in its proposal any pre-existing proprietary rights to be used in its portion of the work, and the license terms for the license of such pre-existing proprietary rights including any reservations or restrictions pertaining thereto. The Proposal shall clearly indicate any areas which the Parties consider to contain pre-existing proprietary rights, and PRIME CONTRACTOR shall be responsible for notifying Customer of any reservations or restrictions pertaining thereto.
- 3.2.7 Each Party shall use reasonable commercial efforts in the preparation of a competitive Proposal and further shall engage in any other reasonable activity which shall result in the acceptance of the Proposal by Customer and the award of the Prime Contract to PRIME CONTRACTOR.
- 3.2.8 If, after submission of the Proposal, Customer requires changes to the Proposal, PRIME CONTRACTOR shall co-ordinate with Subcontractor and submit the response. Subcontractor shall make available at PRIME CONTRACTOR's request employees empowered to make commitments in respect of the matters to be discussed.
- 3.2.9 In the event Customer requires demonstrations or prototypes prior to Subcontractor's execution of a Subcontract, Subcontractor will provide PRIME CONTRACTOR, at mutually agreed charge (if any), with the resources necessary to meet Customer's request.
- 3.2.10 PRIME CONTRACTOR will notify Subcontractor as soon as reasonably possible of the receipt by PRIME CONTRACTOR of the written acceptance or refusal by Customer of the Proposal.

3.3. Subcontract.

- 3.3.1 If PRIME CONTRACTOR is awarded a Prime Contract, the Parties intend to enter into a Subcontract whereby Subcontractor will provide services consistent with the commitments agreed upon in the Proposal. The Subcontract shall be negotiated in good faith between PRIME CONTRACTOR and Subcontractor in advance of or in parallel to the negotiation and execution of the Prime Contract with the understanding that the terms of the Subcontract may need to be modified as a result of subsequent negotiations between PRIME

CONTRACTOR and Customer. The Parties will work in good faith to include any modifications to the Subcontract.

- 3.3.2 The Subcontract shall include substantially similar provisions to those provisions of the Prime Contract which by its terms are required to be flowed down to the Subcontractor.

The Prime Contractor shall make payment to the Company for the services under the Subcontract within **Forty-five (45) calendar days** of receiving the invoice. If payment is not made within this period, the Company reserves the right to terminate the agreement as per Clause 8.3. Additionally, any undisputed amount not paid by the due date will incur interest rate equal to the lower of (i) six percent (6 %) or (ii) Interest rate as per the Repo rate as per RBI per annum from the original due date until payment is made. The Prime Contractor may dispute any invoice within ten (10) business days of receipt; otherwise, the invoice will be considered accepted.

- 3.3.3 In the event a mutually acceptable Subcontract cannot be negotiated and executed by the Parties within a reasonable period of time, and in any event within ninety days (90) days after notice of the award of the Prime Contract, PRIME CONTRACTOR shall have the right upon ten (10) days prior notice to Subcontractor to terminate the relationship and to make other arrangements for the performance of the Work to have been covered by the Subcontract. PRIME CONTRACTOR shall ensure to pay Subcontractor for any pre-agreed costs incurred by the Subcontractor (if any).
- 3.3.4 The Subcontractor understands and agrees that PRIME CONTRACTOR may enter into other arrangements with other individuals or entities under which such individuals or entities would serve as subcontractor to PRIME CONTRACTOR for other products and/or services not part of the Subcontractor's Work under the Proposal.
- 3.3.5 This Agreement shall not be construed as precluding either Party from selling its standard commercially available products to competing bidders.

4 INTELLECTUAL PROPERTY/TRADEMARKS

- 4.1 Subject to section 3.2.6, all intellectual property rights existing prior to the Effective Date of this Agreement shall belong to the Party that owned such rights immediately prior to the Effective Date. Neither Party shall gain by virtue of this Agreement any rights of ownership of copyrights, patents, trade secrets, trademarks or any other intellectual property rights owned by the other.
- 4.2 Neither Party, without the express prior written consent of the other Party, or of the Customer shall use the trademarks, service marks, proprietary words or symbols of the other Party or of the Customer.
- 4.3 Nothing in this Agreement shall affect either Party's right to use any trademarks, service marks or proprietary words or symbols of the other Party to properly identify the goods or services of such other Party to the extent otherwise permitted by applicable law or by written agreement between the Parties.

5. WARRANTY

- 5.1 Company warrants that it has or will have the full power and authority to supply to WIPRO for resale or sublicense to the Customer as the case may be all products, services and other assistance represented by it in the Proposal and will indemnify WIPRO against any actions,

claims, demands, liabilities, damages, costs and expenses arising out of a breach or alleged breach of this warranty or arising out of the Company's intentional or negligent acts or omissions related to the foregoing warranty.

- 5.2 Company warrants that the products, services and other assistance to be provided will be in accordance with the Proposal and will be performed in accordance with the agreed specifications.

6. CONFIDENTIALITY

- 6.1 During the Term of this Agreement, either Party may receive or have access to technical information, as well as information about product plans and strategies, promotions, customers and related non-technical business information which the disclosing Party considers to be confidential ("Confidential Information"). In the event Confidential Information is to be disclosed, the Confidential Information shall be marked as confidential at the time of disclosure, or if disclosed orally but stated to be confidential, shall be designated as confidential in a writing by the disclosing Party summarizing the Confidential Information disclosed and sent to the receiving Party within thirty (30) days after such oral disclosure.
- 6.2 Confidential Information may be used by the receiving Party only with respect to the performance of its obligations under this Agreement, and only by those employees of the receiving Party and its subcontractors who have a need to know such information for purposes related to this Agreement, provided that such subcontractors and employees have signed separate agreements containing substantially similar confidentiality restrictions. The receiving Party shall protect the Confidential Information of the disclosing Party by using the same degree of care (but not less than a reasonable degree of care) to prevent the unauthorized use, dissemination or publication of such Confidential Information, as the receiving Party uses to protect its own confidential information of like nature. Such degree of care shall be no less than be reasonable care.
- 6.3 The obligations stated in this Section 6 shall not apply to any information which is:
- 6.3.1 already known by the receiving Party prior to disclosure.
 - 6.3.2 publicly available through no fault of the receiving Party.
 - 6.3.3 rightfully received from a third Party without a duty of confidentiality.
 - 6.3.4 disclosed by the disclosing Party to a third Party without a duty of confidentiality on such third Party.
 - 6.3.5 independently developed by the receiving Party prior to or independent of the disclosure.
 - 6.3.6 disclosed under operation of law.
 - 6.3.7 disclosed by the receiving Party with the disclosing Party's prior written approval.

7. LIMITATION OF LIABILITY

- 7.1 IN NO EVENT SHALL EITHER PARTY BE LIABLE TO THE OTHER FOR CONSEQUENTIAL, INCIDENTAL OR SPECIAL DAMAGES ARISING FROM ANY CLAIM OR ACTION HEREUNDER, WHETHER BASED ON CONTRACT, TORT OR OTHER LEGAL THEORY.

- 7.2 SUBJECT TO SECTION 7.1 AND NOTWITHSTANDING ANYTHING TO THE CONTRARY ELSEWHERE CONTAINED HEREIN, THE MAXIMUM AGGREGATE LIABILITY OF PARTIES UNDER THIS AGREEMENT SHALL REGARDLESS OF THE FORM OF CLAIM BE LIMITED TO THE AVERAGE ANNUAL FEES PAID OR PAYABLE UNDER THE AGREEMENT.
- 7.3 THE LIMITATIONS IN SECTION 7.2 SHALL NOT APPLY TO (1) BREACH OF SECTION 5.1 BY THE COMPANY (2) BREACH OF CONFIDENTIALITY OBLIGATIONS (excluding personal data as defined under law, which shall be covered by Section 7.2) BY EITHER PARTY; OR (3) THE INDEMNIFICATION OBLIGATIONS OF THE PARTIES.

8. TERM AND TERMINATION

- 8.1 This Agreement shall come into force on the Effective Date and shall continue for a period of twelve (12) months ("Term") unless terminated earlier. The Parties may at any time agree in writing to extend the Term of this Agreement on the same terms and conditions as are contained herein or on such other terms as may be agreed upon by the Parties.
- 8.2 The applicable Work Description shall automatically terminate upon the happening of one of the following events, whichever shall occur first. PRIME CONTRACTOR shall ensure to pay Subcontractor for any pre-agreed costs incurred by the Subcontractor (if any):
- 8.2.1 written notice from a customer that it is cancelling the Request.
 - 8.2.2 written notice from a customer of award of the Prime Contract to a party other than PRIME CONTRACTOR.
 - 8.2.3 notification from Customer stating its disapproval of the use of Subcontractor as a subcontractor or direction to select someone other than Subcontractor for the Work identified as Subcontractor's responsibility in the Proposal or otherwise.
 - 8.2.4 execution of a subcontract by and between the Parties for performance of portions of the Project.
 - 8.2.5 the expiration of the Term of this Agreement; provided, however, this Agreement shall be extended for a period of one additional year if a Proposal has been submitted and Customer has not provided written notice as to the award of the contract within the Term;
 - 8.2.6 the insolvency, bankruptcy, reorganization under the bankruptcy laws, or assignment for the benefit of creditors of either Party;
 - 8.2.7 notice by PRIME CONTRACTOR that it is terminating the Agreement pursuant to Section 3.3.3 above;
 - 8.2.8 mutual agreement of the Parties to terminate the Agreement.
- 8.3 Either Party may terminate this Agreement if the other Party is in material breach of any of its obligations under this Agreement and fails to remedy the breach for a period of thirty (30) days after a written notice by the other Party which specifies the material breach. Either Party may terminate this Agreement without any cause by providing sixty (60) days written notice to the other Party. Both Parties shall ensure to pay the other Party for any pre-agreed costs incurred by the other Party (if any).

9. INDEMNITY

Each party shall indemnify and hold harmless the other party from and against all damages express arising out of or in relation to third party claims in respect of: -of: -

- 9.1. Damage to, or loss of use of, tangible property belonging to the indemnified party caused by the negligent act or omission of the indemnifying party's personnel in connection with performance under this Agreement.
- 9.2. Injury or death of any person, to the extent that such injury or death is caused by the negligent act or omission of the indemnifying party's personnel in connection with the performance under this Agreement.; or
- 9.3. The infringement of a third party's intellectual property rights by any of the products or services provided by the indemnifying party.

10. NON-SOLICITATION AND NON-HIRE

- 10.1 Each Party acknowledges that personnel to be provided by the other Party represent a significant investment in recruitment and training, the loss of which would be detrimental to this Party's business. In consideration of the foregoing, each Party agrees that for the Term of this Agreement and for a period of one year thereafter, either Party will not directly or indirectly, recruit, hire, employ, engage, or discuss employment with any of the other Party's personnel engaged in the performance of this Agreement, or induce any such individual to leave the employ of this other Party. For the purposes of this clause, a personnel means any current person who has worked in such capacity in the previous six months.

11. MISCELLANEOUS

- 11.1 **No Publicity.** Neither Party shall publicize or disclose to any third Party without the consent of the other Party, either the terms of this Agreement or the fact of its existence and execution, except as may be necessary to comply with other obligations stated in this Agreement.
- 11.2 **No Joint Venture.** Nothing contained in this Agreement shall be construed as creating a joint venture, partnership or employment relationship between the Parties hereto, nor shall either Party have the right, power or authority to create any obligation or duty, express or implied, on behalf of the other.
- 11.3 **No Assignment.** Neither Party may assign its rights or obligations hereunder without the other Party's prior written consent, except to the surviving entity in a merger or consolidation in which it participates or to a purchaser of all or substantially all of its assets, so long as such surviving entity or purchaser shall expressly assume in writing the performance of all of the terms of the Agreement. Provided, however, WIPRO may assign this agreement to its affiliates or subsidiaries without the permission of the COMPANY, while remaining responsible to the COMPANY for all payments due to the COMPANY.

- 11.4 **Notices.** Any notice pursuant to this Agreement shall be given in writing and shall be deemed to have been properly given when personally delivered or mailed by certified or registered mail, postage pre-paid, addressed as follows:-

WIPRO : Wipro Limited, Doddakannelli, Sarjapur Road, Bangalore-560035, India.
Representative : generalcounsel.office@wipro.com
Company : nCircle Tech Pvt. Ltd, Unit-2, Office no. 3, 4th Floor, IT -7, Qubix (SEZ), Blue Ridge, Hinjewadi, Phase-1, Pune 411057
Representative : Anand Joshi

- 11.5 **Observance of Regulations.** The employees of each Party shall obey all rules and regulations of the other Party while on the premises of the other Party or the Customer.
- 11.6 **Waiver.** Either Party's failure to exercise any of its rights under this Agreement shall not constitute or be deemed to constitute a waiver or forfeiture of such rights.
- 11.7 **Severability.** If any term or provision of this Agreement is held to be illegal or unenforceable, the validity or enforceability of the remainder of this Agreement shall not be affected.
- 11.8 **Exhibits.** The following document is attached hereto as an exhibit, the terms of which are incorporated by reference in their entirety: Exhibit(s) shall describe Work to be Provided by the Parties based on the Request.
- 11.9 **Precedence.** In the event of conflict between the provisions of this Agreement and any Exhibit, the provisions of this Agreement shall to the extent of such conflict take precedence.
- 11.10 **Survival of Provisions.** Notwithstanding any other provisions of this Agreement, the provisions of paragraphs 3, 4, 5, 6, 7, 9 and 10 shall expressly survive the termination of this Agreement.
- 11.11 **Entire Agreement.** This Agreement and the Exhibits constitute the entire agreement between WIPRO and Company, and supersede any previous or contemporaneous communications, representations or agreements between the Parties, whether oral or written, regarding the subject matter of this Agreement. The terms and conditions of this Agreement may not be changed except by an amendment signed by an authorized representative of each Party.
- 11.12 **Governing Law & Arbitration.**

This agreement shall be governed by and construed in accordance with the laws of India, excluding its conflict of law principles.

Any dispute which cannot be amicably settled by the Parties may be submitted, by one of the Parties, to arbitration under the the Indian Arbitration and Conciliation Act 1996 and any revision thereof. The seat of such Arbitration shall be Pune. The language to be used in the arbitral proceedings shall be English. Both Parties agree that the results and awards of any such arbitration shall (unless a mistake can be shown to have been made in law) be final and binding on both Parties. The Arbitrator shall, be agreed to by the Parties. Failing which, both Parties shall appoint one Arbitrator each and the third Arbitrator shall be decided upon by the

two Arbitrators. Both Parties also agree that the Arbitrator shall have no power to award any exemplary, punitive or indirect damages. Each Party will promptly pay its share of all arbitration fees and costs (provided that such fees and costs shall be recoverable by the prevailing Party as determined by the arbitrator). If a Party fails to pay such share promptly upon demand, the arbitrator shall, upon written request by the other Party, enter a final and binding decision against the non-paying Party for the full amount of such share, together with an award of attorney's fees and costs incurred by the other Party in obtaining such decision, which decision may be entered in any court of competent jurisdiction.

Notwithstanding the determination by the Parties to utilize arbitration as specified above for resolution of disputes arising out of or in connection with this Agreement, nothing herein shall preclude either Party from seeking and obtaining from a court of competent jurisdiction appropriate equitable relief, including without limitation, a temporary restraining order or other injunctive relief, to prevent a breach of this Agreement relating to intellectual property, confidentiality, or non-hire and non-solicitation, or to otherwise maintain the status quo pending outcome of any arbitration.

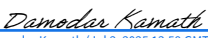
11.13 No person who is not a party to this Agreement shall have any rights to enforce any term of this Agreement.

Contract Reviewed/Executed on behalf of

Wipro Limited

Name: Damodar Kamath

Designation: Account Delivery Executive

Signature/Date: 
Damodar Kamath (Jul 2, 2025 13:59 GMT+5.5)

nCircle Tech Pvt. Ltd

Name: Anand Joshi

Designation: COO

Signature/Date:  Digitally signed by
Anand Waman Joshi
Date: 2025.07.02
10:42:17 +05'30'

Appendix - 1

STATEMENT OF WORK (SOW)

Work order# **WO_25-26_U2_913**

- (A) On **1st July 2025** the Customer and the Supplier entered into Teaming Agreement (the “**Agreement**”).
- (B) The Customer and the Supplier now wish to enter into this Statement of Work which shall incorporate and be subject to the terms of the Agreement.

THE PARTIES AGREE as follows:

1. Definitions and Interpretation

1.1 In this Agreement:

“**Service Commencement Date**” means **1st July 2025**

“**SOW Effective Date**” means **9th June 2025**

“**SOW Expiry Date**” means **8th June 2026**

1.2 Capitalized terms used but not defined in this Statement of Work shall have the meanings given in the Agreement.

2. Governing Law and Jurisdiction

2.1 This Statement of Work shall be governed by and interpreted in accordance with the law of India.

2.2 Any Dispute arising out of or in connection with this Statement of Work shall be resolved in accordance with the Dispute Resolution Procedure set out in the Agreement.

3. Exclusivity

3.1 nCircle agrees that it will not make a competing proposal, whether on a standalone basis or with a third party, to the end customer [**Ericsson**] specifically for the scope of services described in this Statement of Work. Wipro agrees that the foregoing limitation does not prevent nCircle from providing products and/or Services to [**Ericsson**] which fall outside the scope of this Statement of Work.

Project Scope:

1. Project Overview

The main objective of this project is to develop and build a “Scan to BIM” software module to be included in the Ericsson’s Customer Site Digital Twin (ESDT) product. This software module will be Customer IPR.

With this Scan to BIM feature, the following shall be achieved, and this is the most important business requirement:

- IFC files are created based on the Point clouds or 3d meshes, that capture the 3d object shapes, positioning and orientation on site.
- It is intended to be part of the broader Customer digitalization capability for Telecommunications sites as one of the steps of the digitization of telco sites.

Another step in the telco sites digitization process is also required as a distinct feature:

- 3D Shape and dimension recognition is used to do a preliminary identification on the possible equipment items that are on site as this filters the possible items in question.
- This requirement has a lower priority but is included in this Assignment.

The capability required in the assignment are:

- The scope covers in-Shelter and in-Cabinet sites only.
- “Scan to BIM” capability for in-Shelter survey shapes i.e. identify in-Shelter surface shapes within the shelter using a point Cloud or 3D mesh based on a handheld or tripod 360 camera. The output should be IFC.
- “Scan to BIM” capability for Cabinet survey shapes i.e. identify in shelter surface shapes within the cabinet using a pointcloud or 3D mesh based on a handheld or tripod 360 camera towards the open cabinet. The output should be IFC.
- The following are the telecom equipment types are to be identified as part of this assignment
 - Routers
 - Rectifiers
 - Surge Protector
 - XMU's
 - Telco cabinets
 - Cable ladders
 - Microwave IDUs
- The solution will be docker compliant micro services that can be deployed on cloud.
- The deliverable will be directly towards ESDT development and release program including all UAT, integration, deployment and acceptance handling.
- The deliverable must be executable from software orchestration.
- The deliverable must include dimensioning and characteristics definitions for execution once in production e.g. clear definition on GPU requirements, cloud micro services dependencies, 3PP FOSS dependencies and resulting output characteristics.
- Feature instrumentation on DOME calls, execution & creation of IFCs, faults & logging.

2. SOLUTION SCOPE

In-Scope for MVP Phase:

- Solution will consume point cloud data as input and will create the IFC with the following: Structure of the shelter (Walls, entrance), Cabinets, Racks, Cuboids within the racks representing components.
- Integration with ESDT. The solution will consume the files from ESDT, convert to IFC Files and notify the system that automated IFC file is generated.

In-Scope for Post MVP Phase:

- The MVP solution will be improved to include the

- Specific component type identification,
- Details of the specific component,
- Image processing and OCR (Since images are not of good quality)
- The following shapes will be covered in the solution: Routers, Rectifiers, Surge Protector, XMU's, Telco Cabinet, Microwave IDUs.

Out of Scope (Overall):

- Creation of Revit models from point clouds shared.
- Classification of the objects if the exterior dimensions are matching.
- Automation testing of the interface.
- The equipment under Priority 1 and Priority 4 requirement are not considered based on the understanding that Ericsson has a solution for these.
- Reinforced learning pipeline
- Manual Revit Family creation for Telecom components
- Processing on any other data formats like 2d drawing, Survey sheets or legacy models
- FOA Testing, product management, ESDT System test.

What will be delivered in MVP (with the available data of lower quality)

Following items will be delivered as a part of MVP:

- Solution will consume point cloud data as input.
- It will create the IFC with the following:
 - Structure of the shelter (Walls, entrance)
 - Cabinets
 - Racks
 - Cuboids within the racks [represent](#) components. (Only where point cloud data is available)
- Integration with ESDT. The solution will consume the files from ESDT, convert to IFC Files, and notify the system that automated IFC file is generated.

Following items will not be part of MVP:

- Specific component type identification
- Details of the specific component.
- Image processing and OCR (Since images are not of good quality).

Accuracy considerations:

- The accuracy of the cuboids is dependent on the quality of the point cloud data.
- If point cloud of the cabinet is not good quality, the output is likely to have low accuracy.

Definition of done for MVP

- End to end system which Consumes the point cloud data.
- Performs segmentation analysis.
- Performs post processing to convert the segmentation result into geometrical dimensions.
- Generates IFC file.
- Integrated in ESDT
- Performs these operations on 100+ files every week. Scalable solution.
- The MVP will later be enhanced using the higher quality data available later in the project timeline.

Accuracy will not be the criteria to decide the acceptance of the deliverable of MVP.

If good quality data not available?

If the required quality data is not available till MVP

- Accuracy of the output will be impacted.
- In some cases, wrong components dimensions will be detected.
- Data engineering efforts may increase if the quality of the data is not good.
- If new quality data is made available in multiple batches, repeat work to train the ML engine again may increase.
- Tagging data and training ML engine efforts will increase accordingly.

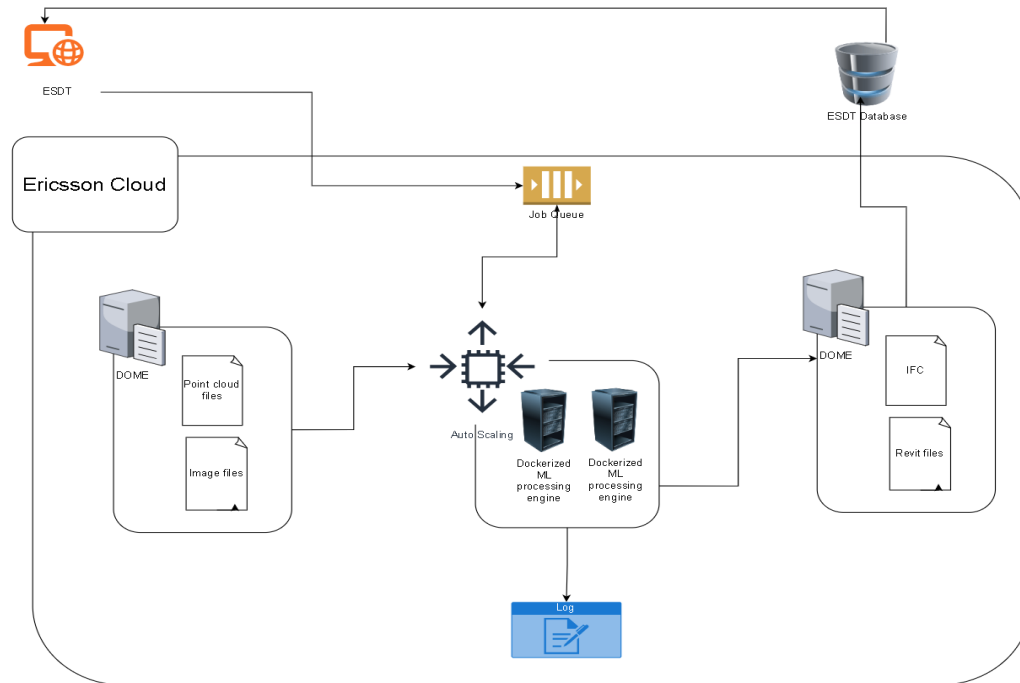
If the required quality data is not available even after MVP

- The acceptance criteria will have to be revised based on the output of the MVP.

Solution Details

1. The solution will develop microservices that will deliver trained machine learning models through the following process
 - Pre-processing: Pre-processing prepares the scan data for further processing. It involves Format Conversion, Property Extraction, Cropping the Point Cloud and Chunking
 - Segmentation stage: This ML-based segmentation stage segments the scan data into meaningful parts. It involves Data Augmentation, Feature Mixing, Sparse Tensor & Convolution, 4D Spatio-Temporal CNN Architecture, Training and Evaluation
 - Post-processing: Post-processing stage refines the segmented data and ensure its accuracy. It involves Data Augmentation, Geometric Modifications and Object Detection Models
 - IFC model creation stage: Dimensional information attained from Post processing stages will be used in creating geometric shapes such as cuboids will be created at the approximate location with approximate dimensions.
 - The solution will be integrated with the ESDT platform and provided a facility where users can select multiple sites for processing and queue to invoke the microservices.
 - A UI screen will be developed where admin can select the sites to be processed or reprocessed
 - The sites will be queued and processed one by one by calling the microservices as required
 - The feedback from the microservices will be stored for reporting purposes

2. High Level Architecture (Indicative)



- The solution will be cloud native and deployable on Azure in Ericsson tenant, with all resource provisioning for environments, build, test, deployment and reporting automated through DevOps
- Single AKS cluster for Scan-to-BIM solution engine and user experience, with a common control plane for operations and control
- Each component service of the solution will be built into a Docker container to be deployed as pods in a cluster
- All components will be custom-built for Ericsson to ensure IP compliance, but will utilize well-established libraries/frameworks available under truly open-source licenses for their internal in order to have realistic implementation timelines
- ESOT is an Azure Cloud hosted application, where most of the components are deployed to an Azure Kubernetes Cluster. To interact with other components HTTP calls or Asynchronous flows with Azure Service Bus is used.

3. Indicative List of Tools & Technologies used by development team:

- NodeJS
- JavaScript
- TypeScript
- Python
- C++
- C#
- Yolo
- Minkowski Engine
- Recap Pro and Recap SDK
- CloudCompare
- VS Code
- Postman

4. Implementation approach

The solution will be delivered with the following steps

- Requirement Evaluation: Evaluating the requirements and analyzing the available data that can be utilized for training and testing.
- Tagging/Labelling of Point Clouds: Preparing the point cloud for training the ML engine.
- Solution Design Phase: Creation of the solution design for the overall solution.
- Development and Unit Testing: Iterative sprint-based development and unit testing of the in-shelter and in-cabinet solutions.
- Solution Integration and Integration testing: Integrating the solution and testing it in the environment
- User Acceptance Testing: User Acceptance Testing to validate the functional and non-functional requirements
- Deployment of the solution on the ESDT platform.
- Go-Live and Post Go-Live Support.

Assumptions & Dependencies

1. Key Assumptions:

- ESDT team will set up development environment of ESDT on developer machine. (We will provide remote access to developer's machine)
- Customer will provide sandbox for the development and integration.
- Team will be using the established logging API of the ESDT portal for their logging in development.
- File storage is provided by ESDT
- File locking not required.
- Server will be high configuration with Windows Server OS
- Development VDI provided by Ericsson for Team should also be of high configuration since we are dealing with point clouds of higher density and size.
- Running cost of ML server during development & training will be borne by Ericsson. Furthermore, this is not included in partner commercials
- Detected top probable components are to be added at the exact position in output IFC which will further be refined by other Ericsson systems
- Sites which do not meet the Input Data Quality Requirements will be out of scope of the solution.

2. Key Dependencies:

Supplier has following dependencies on Customer for successful completion of this assignment:

- User Acceptance Test (UAT) will be done by Customer. Customer will be doing the set up the needed components on the UAT environment and Supplier will be for deploying the required code in UAT environment & provide UAT support to the users
- Technical Support from Customer team through daily sync-up calls, until the project is completed
- Function Support from Customer team on need basis (say, in case of any test findings)

- Permission to perform transfer of scripts, utilities and software modules to Ericsson VDI which are already created by partner(nCircle) for fast tracking the development. Intellectual Property rights will be transferred to Ericsson for such transferred scripts, utilities and software modules.
- Ericsson must provide access to IFC SDK (Discussed & Agreed between both parties) for reading, creating and modifying the input/output IFC file.
- Access to all the environments, systems, and resources needed for the project.
- Processing, storage, infrastructure and technology requirements needed to execute the project including the below:
 - Azure cloud access for server hosting, Azure Kubernetes Cluster and Azure Service Bus
 - Ericsson BIM Object Library
 - Knowledge sharing on Telecom component identification & creation: 6 - 8 Technical Sessions, presuming 1 hour session each
 - Access to Autodesk APS account

3 Pre Requisites

- Revit Licenses: 5 Nos.
- Azure cloud access for server hosting, Azure Kubernetes Cluster and Azure Service Bus
- Ericsson BIM Object Library
- Site Point Clouds for tagging and training: 1000-point cloud sets covering all shapes & types
- Sample BIM or Revit models of shared Point Clouds – 1000 revit files created for the above stated point clouds
- Knowledge sharing on Telecom component identification & creation: 6 - 8 Technical Sessions, presuming 1 hour session each
- ESDT Development environment setup on developer machines
- Access to Autodesk APS account
- The input survey data for multiple sites will be made available within 1st week of the start of the project. This data is recommended to be as per the guidelines shared.

Input Data Quality Requirements (Recommended)

1. Point Cloud

- Ericsson need to mandatorily provide with the following quality available for the program.
- Point cloud:
 - Scanner should be able to capture point clouds with point - point distance ≤ 1 mm
 - Scanner should be taken to the level of the equipment focusing from different possible angles
 - Scanner should be able to capture Point clouds with colours
 - Scanner movement should be slow and steady such that scan will be clear and not blurry
 - Shoulder carry LIDAR scans are preferred so that the details from are that are not in the direct line of sight from tripod.

2. Images

- Images of equipment in the room are captured such that labels on the equipment are at least human readable. The model number and serial numbers should be clear enough for human to be able to read and OCR to work.
- Images should be captured with quality (2048x 2048 pixel per equipment, 4 MB of image data per equipment)
- Images for the equipment near ground level is captured from lower height so that the equipment is clearly visible in the images.
- Panoramic images are preferred.
- Images of the equipment rack should be such that one equipment rack has one clear image from front.
- IF there are equipment in the cabinets, the door of the cabinets should be opened, and the images should be captured in good lighting.
- Scanning team to ensure that there is minimal obstruction of any kind on the labels and the equipment when taking the photos
- If photos are taken from another device, the angle orientation of the images should be captured so that we can position the photo in 3D space.
- Point cloud data and respective IFC/Revit file data available for creating automatic tagging script.

MVP Approach

- Development team will start the development with available low quality of point cloud data.
- Ericsson team will arrange to transfer 1000+ site low quality data to development team.
- Development team will train the ML engine on the low-quality point cloud data.
- As and when better quality point cloud data is available, the new data will be ingested into the ML engine to generate better results.
- It was also agreed that the accuracy that can be achieved using this automation will be defined during the implementation phase as a joint effort with Ericsson team.

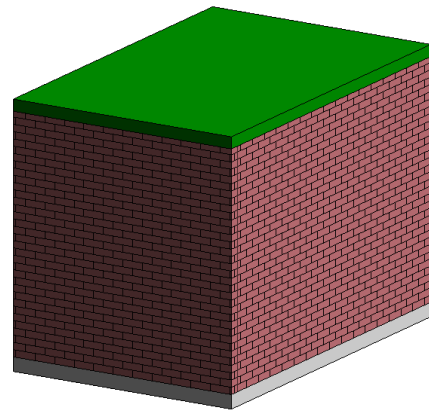
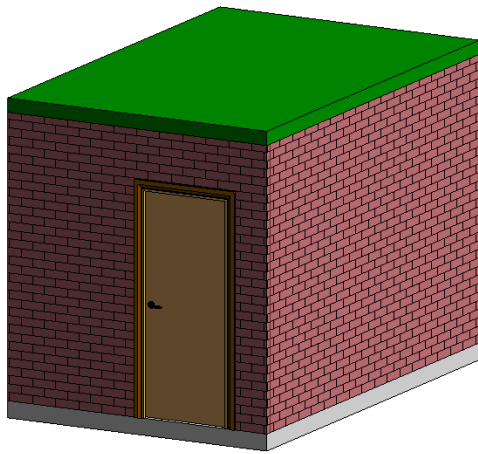
Acceptance Criteria OR Definition of Done for MVP

The MVP will be considered completed when End to end system which -

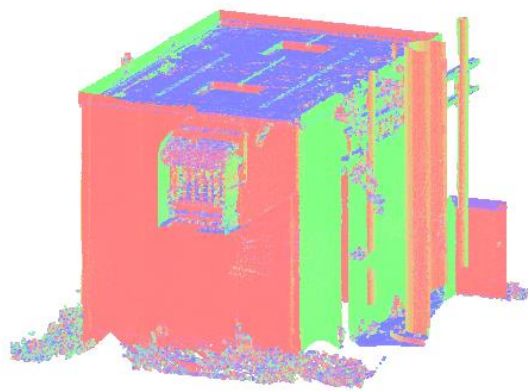
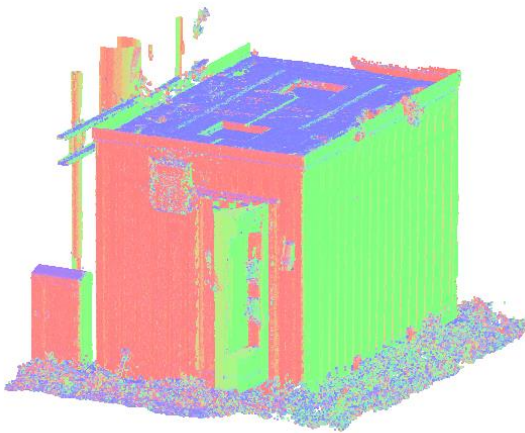
- Consumes the point cloud data.
- Performs segmentation analysis.
- Performs post processing to convert the segmentation result into geometrical dimensions.
- Generates IFC file.
- Integrated in ESDT
- Performs these operations on 100+ files every week. Scalable solution.
- The MVP will later be enhanced using the higher quality data available later in the project timeline.
- Accuracy will not be the criteria to decide the acceptance on the deliverable of MVP.

Below are the reference images of expected output as part of MVP delivery

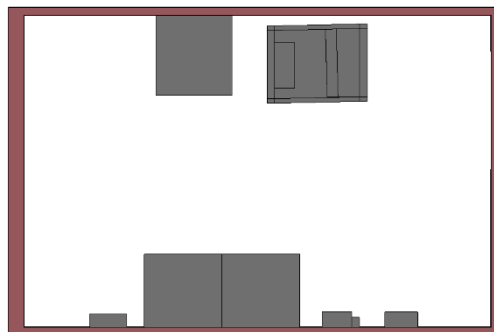
REVIT MODEL - VIEWS



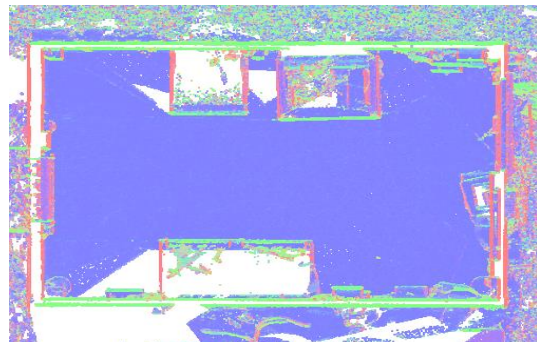
POINT CLOUD DATA – VIEWS (Normals views)

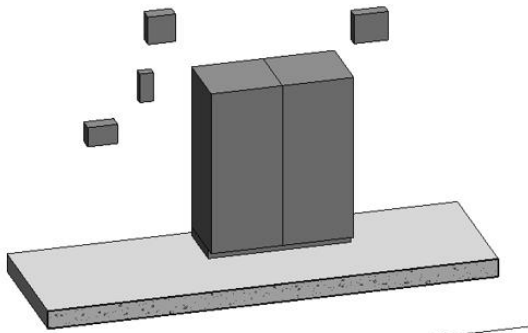


REVIT MODEL - FLOOR PLAN VIEW

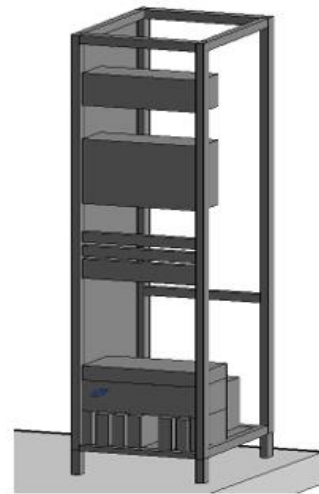


POINT CLOUD DATA - FLOOR PLAN VIEW

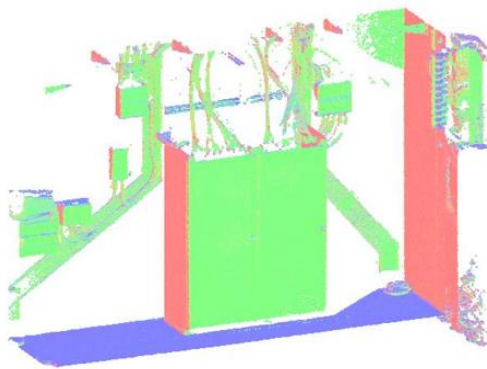




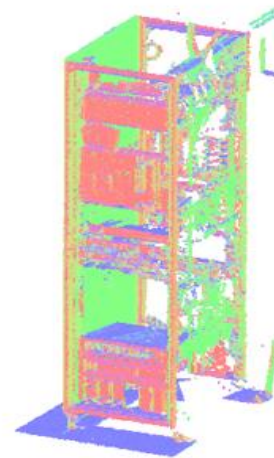
REVIT MODEL – VIEW



REVIT MODEL – VIEW



POINT CLOUD DATA - VIEW



POINT CLOUD DATA - VIEW

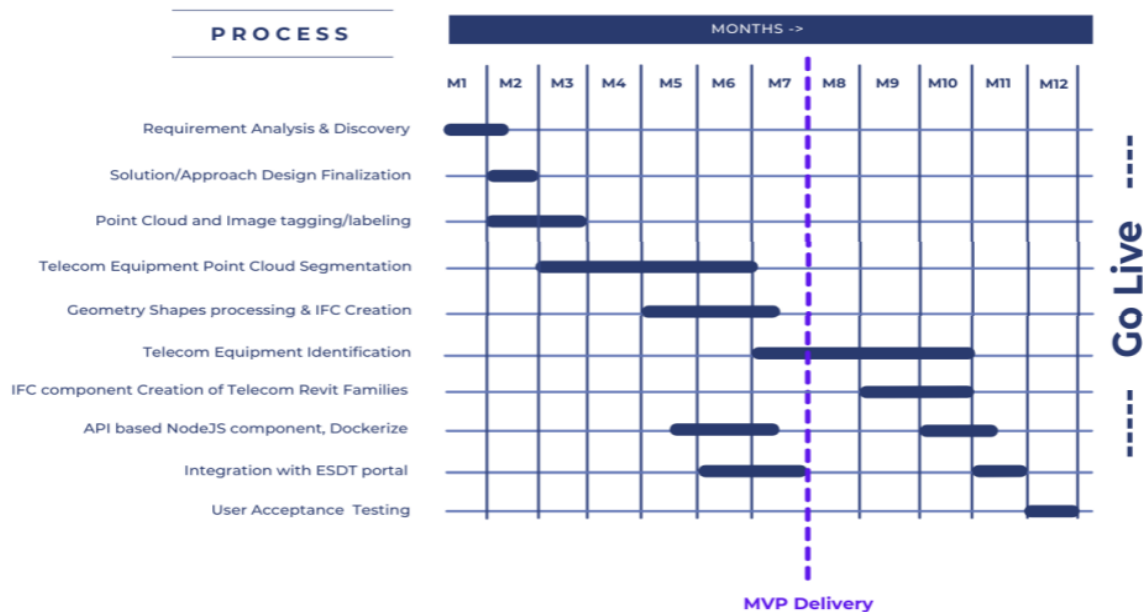
Key Deliverables:

Item	Deliverable Description	Acceptance Criteria
1.	Analysis and Feasibility Report	Completed analysis report with recommendation on feasibility of using data for further processing.
2.	Solution Design	Review of the solution design with Ericson stakeholders
3.	Code and UT Results	All code deployed in Ericson UAT environment.
4.	Release Verification	Successful deployment of microservices and integration. Successful release verification for users to verify functionality of IFC creation for in-shelter shapes in input point cloud scans.

5.	Release to production environment	Successful release to production. No outstanding critical issues** at the end of 1 month after the release. **Any issue which restricts creation of output IFC file is considered a critical issue. Output quality-related issues are not considered a critical issue.
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Project Schedule:

Ericsson Automated Scan To BIM Timeline



* This timeline created assuming 1st June as start date and first usable ESDT integration is expected by end 2025

We propose a checkpoint after the MVP to decide on the next steps

RASCI Responsibility Matrix:

Subprocess/Activity/Task	Responsible	Accountable	Supporting	Consult	Inform
Arrange the data	Ericsson	Ericsson	Wipro	Partner (nCircle)	Partner (nCircle)
ESDT Development environment setup	Ericsson	Ericsson	Wipro	Partner (nCircle)	Partner (nCircle)
Development of the Core module microservices	Partner (nCircle)	Partner (nCircle)	Wipro	Ericsson	Ericsson
Development of the Queuing function and integration	Wipro	Wipro	Partner (nCircle)	Ericsson	Ericsson

Deployment	Wipro	Wipro	Partner (nCircle)	Ericsson	Ericsson
Project management and User stories detailing	Wipro/ Partner (nCircle)	Wipro	Partner (nCircle)	Ericsson	Ericsson

Indicative List of Technology Stack:

- **Web** - Node.js, Jest, TypeScript, Azure (i.e. App Services, Azure service Bus, Kubernetes Cluster, Monitor, container registry etc.)
- .NET, Python, NumPy, Pandas, SciPy, matplotlib, Open 3D, Scikit-learn, OpenCV, pillow, ply file, pye57, C++ (optional).
- spconv, MinkowskiEngine , torch-scatter, torch_sparse, torch_points3d, torch_geometric, torch_points_kernels, Segmentator, pointnet2_ops, Pytorch , MKL, PDAL.
- Db-PostgreSQL.
- **Tools** – Git or Azure DevOps or any other repository access, VSCode/Pycharm , Docker desktop, Visual Studio , pgAdmin, cloud compare, Azure CLI, Helm, GCC/G++, CMake.

Please Note: Any additional VDI requests by nCircle team (if required), shall be accommodated by Wipro/Ericsson.

Escalation Matrix:

For any kind of escalation related to work, billing, or communication it should be raised via email communication to the below mentioned parties:

- o **First Level Escalation:** Ankita Badoni (Project Lead) (ankita.badhoni@ncircletech.com) OR Pavan Sikakolli (pavan.sikakolli@ncircletech.com)
- o **Second Level Escalation:** Anirban Sarkar (anirban.sarkar@ncircletech.com)
- o **Third Level Escalation:** Anand Joshi (COO) (anand.joshi@ncircletech.com)

Charges

1. **Engagement Type** – Fixed Capacity Progressive Billing – **USD 280,000 + Taxes if any.**
2. **Milestone and Payment Schedule -**
Payment Terms:
 - a. nCircle will raise an invoice at the end of each month as described along with the associated the amount mentioned in Appendix 2.
 - b. Customer agrees pay each invoice in NET 45 days from date on invoice.

Please Note: if invoice is raised to Wipro India entity (outside SEZ) IGST of 18% will be applicable additional to the below price.

Month	Brief description	Monthly invoice in USD (Blended rate of team)
Jun-25	Discovery sessions, Input Data Analysis	15000
Jul-25	Solution design, Automated Point Cloud Tagging Scripts	20000
Aug-25	Point Cloud Tagging, ML Segmentation Model Creation	20000
Sep-25	MVP - ML Segmentation Model Configuration, Training	25000
Oct-25	MVP - ML Segmentation Model Optimization, Training, Geometry Shape Detection	25000
Nov-25	MVP - ML Segmentation Output Refinement, IFC component creation, ESDT Integration, UAT	25000
Dec-25	MVP - IFC file completion, Complete ESDT integration, UAT, Telecom Component Identification ML Engine Creation	25000
Jan-26	Telecom Component Identification ML Engine configuration, training	25000
Feb-26	Telecom Component Identification ML Engine Optimization, Match identified component with Ericsson Component Library	25000
Mar-26	ML Identification Output Refinement, NodeJS component creation for Telecom Identification	25000
Apr-26	Dockerise Identification NodeJS comp, ESDT platform integration	25000
May-26	User Acceptance Testing	15000
Jun-25	Pipeline maintenance, Bug Fixing, Documentation	10000
	Total	280000

- Supplier shall provide the Services in accordance with the milestones and dates shown above. Supplier will only invoice upon Customer's acceptance of all the corresponding deliverables associated with the milestone payment. Acceptance of a delivery at a milestone will be made according to the provisions above.

- Reimbursable Expenses.** Customer shall reimburse Supplier for pre-authorized reasonable expenses incurred under this SOW, if expressly set forth in and in accordance with the Agreement. If applicable, Supplier shall invoice Customer for such actual substantiated expenses in accordance with the invoicing procedures in the Agreement.

- Customer's billing details:**

Contact Person: [edebtors-europe](#)

Email: edebtors-europe@wipro.com

Billing Address: Wipro Limited, SEZ, Unit III, Sy. No. 91/1, 98 & 99, Kodathi Village, Varthur Hobli, Sarjapur Road, Bangalore East Taluk, Bangalore -560035, India, ("Wipro").

- All licenses required to carry out the development activities would be provided by Customer.

7. **Invoices and Payment:** The Prime Contractor shall make payment to the Company for the services under the Subcontract within Forty five **(45) calendar days** of receiving the invoice. If payment is not made within this period, the Company reserves the right to terminate the agreement as per Clause 8.3. Additionally, any undisputed amount not paid by the due date will incur interest rate equal to the lower of (i) six percent (6 %) or (ii) Interest rate as per the Repo rate as per RBI per annum from the original due date until payment is made. The Prime Contractor may dispute any invoice within ten (10) business days of receipt; otherwise, the invoice will be considered accepted.

General Terms and Conditions

For holidays, the resources will follow the holiday list of nCircle Tech (the screenshot is mentioned below for 2025-2026). The holiday list for 2026-2027 would be shared with the client by March 2026.

Holiday List 2025-2026			
Sr.No.	Occasion	Date	Day
1	Maharashtra Day/ International Labour Day	1-May-25	Thursday
2	Independence Day	15-Aug-25	Friday
3	Ganesh Chaturthi	27-Aug-25	Wednesday
4	Dushhehra	2-Oct-25	Thursday
5	Diwali- Laxmi Pujan	21-Oct-25	Tuesday
6	Diwali-Padwa	22-Oct-25	Wednesday
7	Christmas	25-Dec-25	Thursday
8	New Year Day	1-Jan-26	Thursday
9	Dhulivandan	3-Mar-26	Tuesday

Optional Holiday			
Sr.No.	Occasion	Date	Day
10	Makar Sankranti	14-Jan-26	Wednesday
	Ramzan Eid	20-Mar-26	Friday

- Each Developer is allowed to take 2 leaves per month. Leaves will be informed in advance unless sick leaves (provided, however, that in the case of sick leaves, in advance or if it is an emergency, as soon as possible). In case of additional days off days beyond 2 days per month, the Supplier shall make a written notice to the customer.

Warranty: Supplier warrants that the services will be performed in a good and workmanlike manner. EXCEPT AS SPECIFIED HEREIN, ALL EXPRESS OR IMPLIED CONDITIONS, REPRESENTATIONS AND WARRANTIES, INCLUDING ANY IMPLIED WARRANTY OR CONDITION OF MERCHANTABILITY, SATISFACTORY QUALITY, FITNESS FOR A PARTICULAR PURPOSE ARE HEREBY DISCLAIMED. Customer's sole and exclusive remedy and Supplier's entire liability for breach of the above warranty for a Deliverable will be to rectify or replace such defective deliverable. Any claim for breach of Supplier's warranty must be provided to Supplier in writing within 30 days of performance of the Services or delivery of Deliverable at issue.

Compliance with Laws: Each Party will retain responsibility for its compliance with all applicable laws and regulations relating to its respective business and facilities and the provision of services to third parties, including applicable industry specific laws, regulations

and orders. In performing their respective obligations under this SOW, neither Party will be required to undertake any activity that would violate any applicable laws or regulations.

END OF DOCUMENT